

KYLE THOMPSON

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EDUCATION

Northwestern University

Expected Dec 2026

Master of Science in Robotics

University of Illinois at Urbana-Champaign

Aug 2020 - Dec 2024

Bachelor of Science in Mechanical Engineering, Minor in Computer Science

TECHNICAL SKILLS

Software Development: C, C++, Python, Java, MATLAB/Simulink, Bash, Git, Linux, Docker, Unit Testing, CI/CD, Pandas

Robotics: ROS/ROS 2, Pytorch, OpenCV, Path Planning, Kinematics, MoveIt2, SLAM, HuggingFace, IsaacSim, Nav2

Hardware: LiDAR, IMUs, RGB-D, Nvidia Jetson, 3D-printing, PLM ECO, Rapid Prototyping, NX, SolidWorks

WORK EXPERIENCE

Vermeer

Brooklyn, New York

Robotics Software Engineering Intern

Jun 2026 – Current

- Optimized FR-20 cobot camera calibration for high-precision computer vision pipeline to decrease cycle time by 50%
- Integrated robot trajectory spline via frame transforms to reduce camera-IMU calibration reprojection error by 40%
- Developed python integration test framework for calibration system, increased calibration success rate by >100%
- Deployed containerized machine learning inference on Nvidia Jetson Orin for EKF covariance scaling analysis

Willow

Mountain View, CA

Engineering Intern

Apr 2025 – Aug 2025

- Prototyped multi-antenna Bluetooth Channel Sounding distance measurement embedded firmware with C
- Integrated Zephyr RTOS, UART communication protocols, multithread synchronization in proof-of-concept system
- Developed Python data analysis tool to compare vacuum performance across 2,000 pumps
- Performed DFU and System validation of embedded firmware on 150 Willow Sync pumps

Mechanical Engineering Intern

Jun 2024 – Aug 2024

- Applied NX and SLA printing to rapidly fabricate soft plastic prototypes in design sprint to product release
- Created test models and generated vacuum waveforms to improve prototype system efficiency by 20%

Ancora Heart

Santa Clara, CA

R&D Intern

Aug 2023 – Dec 2023

- Executed verification testing, fixture development and data analysis to support catheter performance evaluation
- Assembled, programmed C, and installed automated catheter braiding system to reduce braiding time by 50%

PROJECT EXPERIENCE

Data Driven World Model on Robot Arm (ROS 2, Pytorch)

Mar 2026 – Current

- Integrated MPC and RSSM exploring World Model on Widow x200 arm to learn environment in from camera feed
- Experimented with sequenced generated goal images and DINO segmentation to guide robot manipulation behavior

Language Guided SLAM on Unitree Go (ROS 2, Python, C++)

Jan 2026 – Mar 2026

- Built semantic memory concept graph implementation with SAM3 and CLIP through live remote GPU inference
- Applied RTAB-MAP, Nav2, sensor fusion, and nearest frontiers on Go2 for SLAM and autonomous exploration

Simultaneous Localization and Mapping (SLAM) from Scratch (ROS 2, C++)

Jan 2026 – Mar 2026

- Developing an Extended Kalman Filter SLAM pipeline library for use on a differential drive wheeled turtlebot
- Implemented a landmark detection algorithm using supervised learning and data association for state estimation

End-to-End Optics and Event Stream Pipeline (Pytorch)

Jan 2026 – Mar 2026

- Designed an end-to-end event-stream to spiking neural network pipeline in PyTorch for optical flow
- Generated synthetic optic flow datasets in Isaac Sim for GPU inference evaluation of pretrained vision models

Ball Catching Franka Arm (Python, ROS 2, YOLO)

Nov 2025 – Dec 2025

- Led team of 4 engineers using Git to develop motion planning ROS 2 package with MoveIt2 on Franka Emika Panda
- Trained YOLOv11 object detection model with 5000 image dataset to identify moving balls for trajectory prediction
- Conducted eye-in-hand camera calibration using RGB-D camera and ArUco markers for precise 3D localization